Deformation Theory of Calabi-Yau Varieties

Midterm Defense

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Outline

1 Preliminaries

2 Complex Geometric Approach

3 Algebraic Approach

4 Expected Outcomes

Convention

- Throughout, $k = \mathbb{C}$ (algebraically closed field of characteristic 0).
- Varieties are assumed to be projective, so by GAGA we can freely switch between algebraic and analytic viewpoints.
- A **Calabi-Yau manifold** is a compact Kähler manifold X with $K_X \cong \mathcal{O}_X$ (trivial canonical bundle).

Two Approaches to Deformation Theory

Complex Geometric

- Complex structure = integrable almost complex structure.
- Deformation encoded by $\phi(t) \in \mathcal{A}^{0,1}(T_X)$.
- Integrability $\Longleftrightarrow$ Maurer-Cartan equation:

$$\bar{\partial}\phi + \tfrac{1}{2}[\phi, \phi] = 0.$$

- Kodaira-Spencer class $[\phi_1] \in H^1(X, T_X)$.

Algebraic

- Deformation functor $\mathrm{Def}_X : \mathrm{C} \to \mathrm{Set}$.
- C = local Artinian k-algebras with residue k.
- Obstruction theory via T^i functors (Lichtenbaum–Schlessinger).
- Schlessinger's criterion for pro-representability.

Maurer-Cartan Equation

Let $X = (M, I)$ be a complex manifold. A deformation $I(t)$ of I gives a map

$$\phi(t) : T^{0,1} \to T^{1,0}, \quad \phi(t) = -\operatorname{pr}_{T_t^{1,0}} \circ j.$$

Lemma

Integrability $[T_t^{0,1}, T_t^{0,1}] \subset T_t^{0,1}$ *is equivalent to*

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Expanding $\phi(t) = \sum_{i \geq 1} \phi_i t^i$ yields a recursive system:

$$\bar{\partial}\phi_1 = 0, \quad \bar{\partial}\phi_2 + [\phi_1, \phi_1] = 0, \quad \dots$$

Hence first-order deformations correspond to $H^1(X, T_X)$.

Tian–Todorov Lemma

On a Calabi-Yau manifold, fix a holomorphic volume form Ω. Define an isomorphism

$$\eta : \bigwedge^{p} \mathcal{T}_X \xrightarrow{\cong} \bigwedge^{n-p} \Omega_X, \quad \eta(v_1 \wedge \cdots \wedge v_p) = i_{v_1} \cdots i_{v_p}(\Omega).$$

Lemma (Tian–Todorov)

For $\alpha \in \mathcal{A}^{0,p}(\mathcal{T}_X)$, $\beta \in \mathcal{A}^{0,q}(\mathcal{T}_X)$,

$$(-1)^p[\alpha, \beta] = \Delta(\alpha \wedge \beta) - \Delta(\alpha) \wedge \beta - (-1)^{p+1}\alpha \wedge \Delta(\beta),$$

where $\Delta = \eta^{-1}\partial\eta$ anti-commutes with $\bar{\partial}$.

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Consequence: If $\eta(\phi_1)$ is harmonic, then $[\phi_1, \phi_1]$ is $\bar{\partial}$-exact, allowing recursive solution of MC equation.

Unobstructedness for Smooth CY

Theorem (Bogomolov–Tian–Todorov)

Let X be a smooth Calabi-Yau manifold. Then the deformation functor is unobstructed: every first-order deformation extends to a formal (and convergent) deformation.

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Key steps:

1. Choose harmonic representative ϕ_1 using Hodge theory.
2. Inductively solve $\bar{\partial}\phi_k = -\sum_{0<i\leq k}[\phi_i, \phi_{k-i}]$ using the Tian–Todorov lemma and the $\partial\bar{\partial}$-lemma.
3. The formal solution converges by analytic estimates (Kodaira–Nirenberg–Spencer).

The T^i Functors (Lichtenbaum–Schlessinger)

For a ring homomorphism $A \to B$ and a B-module M, define $T^i(B/A, M)$ via the cotangent complex:

$$L_2 \xrightarrow{d_2} L_1 \xrightarrow{d_1} L_0,$$

where $L_0 = \Omega_{R/A} \otimes_R B$, $L_1 = F \otimes_R B$, $L_2 = Q/F_0$ from a presentation $R = A[x] \twoheadrightarrow B$ and $F \twoheadrightarrow I = \ker(R \to B)$.

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- $T^0(B/A, M) = \mathrm{Der}_A(B, M) = \mathrm{Hom}_B(\Omega_{B/A}, M)$.
- For $B = A/I$, $T^1(B/A, M) = \mathrm{Hom}_B(I/I^2, M)$ (normal module).
- Smoothness $\iff$ $T^1(B/k, M) = 0$ for all M.

Schlessinger's Criterion

Let $F : \mathsf{C} \to \mathsf{Set}$ be a functor of Artin rings.

Theorem (Schlessinger)

*F has a **miniversal (versal) family** iff:*

(H0) *$F(k)$ is a singleton.*

(H1) *$F(A' \times_A A'') \to F(A') \times_{F(A)} F(A'')$ surjective for $A'' \to A$ small.*

(H2) *The map is bijective when $A'' = D$, $A = k$.*

(H3) *$t_F = F(D)$ is a finite-dimensional k-vector space.*

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For projective X, Def_X satisfies (H0)-(H3); pro-representability holds when $\mathrm{Aut}(X/X_0)$ lifts.

Ran–Kawamata T^1 Lifting

Definition

$T^1(X_n/A_n) := \mathrm{Def}(X_n/A_n, B_n)$ where $A_n = k[t]/(t^{n+1})$, $B_n = A_n \otimes A_1$.

Theorem (Kawamata, 1992)

Assume Def_X *is pro-representable and satisfies the* T^1 ***lifting property****: for every* $X_{n+1} \in F(A_{n+1})$,

$$T^1(X_{n+1}/A_{n+1}) \to T^1(X_n/A_n) \quad \textit{is surjective.}$$

Then F *is unobstructed (the base ring is a power series ring).*

Ran–Kawamata T^1 Lifting

The T^1 lifting property for smooth CY follows from Hodge–de Rham spectral sequence degeneration and the fact that $H^q(X_n, \Omega^p_{X_n/A_n})$ are free A_n-modules.

Work Completed and Planned

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- Maurer-Cartan equation and Tian–Todorov lemma.
- Algebraic deformation theory: T^i functors, Schlessinger's criterion.
- Ran–Kawamata T^1 lifting and proof of unobstructedness for smooth CY.

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Ongoing / Future:

- Prepare a final thesis summarizing deformation theory of CY varieties from smooth to mildly singular cases.

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Thank you!

Questions and comments are welcome.